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Review

Overcoming procrastination? A meta-analysis of intervention studies

Wendelien van Eerde[a,*], Katrin B. Klingsieck[b]

[a] University of Amsterdam, the Netherlands
[b] University of Paderborn, Germany

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ABSTRACT

We present a meta-analysis of 24 studies on procrastination interventions (total $k = 44$, $N = 1173$) in order to find out 1) whether people can reduce their level of procrastination, and 2) if so, which type of intervention leads to the strongest reduction. We compared four different types of interventions: Self-regulation, cognitive behavioral therapy, other therapeutic approaches, and interventions focusing on individuals' strengths and resources. A large reduction in procrastination after the interventions was found, and the effects remained stable in follow-up assessments. The findings so far suggest that cognitive behavioral therapy reduced procrastination more strongly than the other types of interventions. Other moderator variables, such as the duration of the intervention, had no significant effects. We propose future research that may help to build stronger evidence for the effects of interventions, as well as some guidelines for interventions.

1. Introduction

Have you ever delayed something that you thought should be done right away, even though you knew that delaying it would be bad? If so, you have engaged in irrational delay, or procrastination. Most people recognize it as something they do occasionally. However, procrastination may also be a chronic tendency a habit of needless delaying things that need to be done (Ferrari, Diaz-Morales, O'Callaghan, Diaz, & Argumedo, 2007). If this is the case, chances are that you know it is difficult to get rid of, even if you wish to stop behaving like this. Particularly among students, chronic procrastination appears to be problematic. Many studies have shown that procrastination affects students' grades and well-being; procrastination may have severe consequences for academic achievement (Kim & Seo, 2015). Because the vast majority of the studies on procrastination focuses on academic procrastination, i.e., procrastination of study-relevant activities of university students, the following discussion of this phenomenon is, strictly speaking, a discussion of academic procrastination.

Many studies have been conducted on procrastination. Several meta-analyses have combined the effect sizes of the studies measuring procrastination in relation to other psychological variables, each with a somewhat different focus. The earlier meta-analyses (Steel, 2007; Van Eerde, 2003a, 2003b; van Eerde, 2004) provided a general overview of how procrastination is related to many variables, such as personality, outcomes in terms of academic performance, and mental health. More recent meta-analyses have addressed specific topics, focusing on the relation between procrastination and time perspective (Sirois, 2014), coping (Sirois & Kitner, 2015), and academic performance (Kim & Seo, 2015). Overall, these meta-analyses show that procrastination is detrimental to individuals' health and well-being, as well as their achievements. As such, it is important to refine our knowledge on procrastination interventions.

* Corresponding author.0
E-mail address: w.vaneerde@uva.nl (W. van Eerde).

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An important debate surrounding the topic of procrastination concerns its stability. Can people change this behavior? On the one hand, the research indicates that it is stable, at least some may be attributed to a genetic component (Gustavson et al., 2017; Gustavson, Miyake, Hewitt, & Friedman, 2014), and that there are strong relations with relatively stable variables, such as conscientiousness and impulsivity (Steel, 2007). On the other hand, the research also suggests that over the course of a lifetime people may change their procrastination behavior. Procrastination is negatively related to people's age, and one specific group, i.e., men under thirty, appear to procrastinate most (Beutel et al., 2016). Another indication that procrastination may be overcome is that many self-help books have been written on how to tackle procrastination (e.g., Burka & Yuen, 2008). Several interventions to overcome procrastination have been developed. Not all have been based on research evidence, but some studies have been devoted to the effects of these interventions using systematic and scientific approaches.

The current study focuses on these intervention studies. We integrated the effects of these studies meta-analytically to assess whether, and if so to what extent, procrastination decreases after an intervention. It is the first attempt of systematizing and synthesizing the effect sizes of procrastination interventions. This systematization enables us to compare studies in order to establish whether some types of interventions (e.g., therapy) obtained larger effects than others. It also enables us to assess if, and if so to what extent, changes after an intervention endure. That is, whether studies indicate that changes may last up to a certain period following the intervention.

With this meta-analysis, we aim to contribute to the literature by providing an average effect size of quantitative intervention studies on procrastination. By doing so, we offer a tentative answer to the question whether interventions actually help to reduce procrastination. Our second contribution is to provide insight into the relative effectiveness of the different interventions by focusing on their design characteristics in terms of content and application. This might help counselors, coaches, and others who aim to help procrastinators. Our findings offer implications for research on and guidance of those who suffer from procrastination by summarizing previous work and pointing out which approaches appear to be most promising for the future. Our analysis may also indicate whether certain interventions may not appear to be worthwhile pursuing, possibly saving time and effort for all those involved.

In the following, we will first discuss procrastination in more detail, elaborating on how it has been conceptualized. We will then provide an overview of what has been done so far to overcome it, and how procrastination may be measured. Subsequently, we will present our meta-analysis.

1.1. Theoretical background

Delay may be rational, strategic, and beneficial, when it is a conscious choice to postpone activities under the control of the individual. However, procrastination is an irrational and acratic behavior, since it “is to voluntarily delay an intended course of action despite expecting to be worse off for the delay” (Steel, 2007, p. 66). Often people indicate that they cannot control their procrastination. They cannot help but do it, finding themselves doing it again even if they would like to start right away (Klingsieck, 2013a; Van Eerde, 2000). How is this possible?

Procrastination is widely acknowledged as a self-regulation failure, if not as the “quintessential self-regulation failure” (Steel, 2007). Self-regulation encompasses the processes needed to deploy our cognitive, emotional, and behavioral resources in order to reach a desired goal or outcome (Baumeister & Heatherton, 1996), similar to a CEO managing resources of her/his company in order to reach a goal. As failure of self-regulation, procrastination is associated with problems in all three phases of self-regulation (Zimmerman, 2000): In the pre-actional phase procrastination is related to missing self-determination concerning the task at hand (e.g., Senécal, Julien, & Guay, 2003) and associated with problems in planning and prioritizing tasks (e.g., Lay & Schouwenburg, 1993). In the actional phase, it is associated with problems in concentrating on the task and shielding distractions (e.g., Dewitte & Schouwenburg, 2002), while in the post actional phase it is associated with low self-efficacy (e.g., Wäschle, Allgaier, Lachner, Fink, & Nückles, 2014) which then determines the type of self-motivation for the next pre-actional phase. Specifically, academic procrastination seems to be related to a deficit use of cognitive and meta-cognitive learning strategies (e.g., Howell & Watson, 2007).

One can be either consciously aware of certain techniques or the processes run more or less unconsciously. Kuhl (1996) distinguished the term “self-regulation” for the unconscious form from the term “self-control” for the conscious form of regulating oneself. Self-control is especially important when long-term goals need to be achieved that might entail the abdication of short-term gratification. It is a global, domain non-specific resource whose capacity is assumed to be finite (Baumeister, Bratslavsky, Muraven, & Tice, 1998). In procrastination research, the terms *self-regulation* and *self-control* are used synonymously when procrastination is conceptualized as self-regulation failure, thus, when we use the term *self-regulation* it encompassed the unconscious and the conscious form.

While procrastination is a self-regulation *failure,* responsible for not achieving goals, strategic delay can be seen as a self-regulation *strategy*, helpful in achieving goals. Lately, authors have begun to explicitly acknowledge the difference between procrastination as self-regulation failure and other forms of delay as self-regulation strategies (Corkin, Yu, & Lindt, 2011; Grunschel, Patrzek, & Fries, 2013; Klingsieck, 2013a; Krause & Freund, 2014). Only very few studies investigate both the self-regulation failure and self-regulation strategy, using the same term “procrastination”. Chu and Choi (2005) distinguish procrastination and active procrastination. However, this approach had been criticized on both theoretical and empirical grounds (Chowdhury & Pychyl, 2018).

Superficially, procrastination and strategic delay look alike. However, the two forms of delay have different cognitive, motivational, and emotional effects. Procrastination is accompanied by some degree of subjective discomfort, such as fear of failure (Haghbin, McCaffrey, & Pychyl, 2012). The discomfort may lead to either externalizing or internalizing types of behavior (Gustavson et al., 2017). Many find procrastination emotionally burdening, even though the distraction and emotion regulation through other activities may provide temporary relief (Sirois & Pychyl, 2013; Van Eerde, 2000). Procrastination hampers the achievement of goals, is associated with feelings of guilt and shame, and is experienced as stressful (Myrick, 2015; Reinecke, Hartmann, & Eden, 2014;

Sirois, 2013). Empirical findings show that self-reported procrastination is related to but not the same as dilatory behavior or missing a deadline (Van Eerde, 2003a, 2003b). By contrast, strategic delay entails the conscious knowledge that the positive consequences of the delay will outweigh the potential negative consequences of the delay (Klingsieck, 2013a).

In addition to the affective ill-being associated with procrastination, the subjective appraisal of a delay is important because internal norms play a role (Milgram & Naaman, 1996; Van Eerde, 2000). Procrastination cannot be observed directly by others: If others think that the delay is within an acceptable range, but the individual does not think so, it may still be problematic to the individual. Individuals may differ with respect to what they find an acceptable delay in comparison to the social norm. Also, norms on what is acceptable delay may differ between cultures (White, Valk, & Dialmy, 2011), and these norms may also differ between settings or life domains within a culture (Klingsieck, 2013b). When individuals consider the delay as acceptable but others do not, there is also a discrepancy between the internal norm and the social norm, but this does not appear to be within the phenomenon studied so far. Thus, measuring procrastination always entails a subjective flavor.

Conceptually, it is imaginable that people procrastinate in only one life domain, and not in other domains. For example, a student may procrastinate in her studies but does not procrastinate in her household. However, the domains cannot be strictly separated empirically (Klingsieck, 2013b). If students procrastinate in the academic domain, it is highly likely they procrastinate in other domains as well, and vice versa. However, most studies have focused on students in the educational domain because self-regulation failures, such as procrastination, are detrimental to academic achievement (cf. Kim & Seo, 2015). Thus, with their focus on factors that are crucial to academic achievement such as motivation, self-regulation, and learning strategies self-regulation failure interventions are very relevant to education.

Summing up, in the case of procrastination, there is an intention-action gap rather than an intention to delay. Particularly in light of behavioral change, it is important to understand that procrastination is irrational and self-defeating. Many who procrastinate would like to change their behavior but find it difficult to do so (cf. Grunschel & Schopenhauer, 2015). As such, we agree with Haghbin's (2015) statement that it is important to take into account whether procrastination is causing discomfort, and whether an individual has experienced discomfort over longer periods in the past. Given this discomfort and the difficulties experienced in overcoming procrastination, it is no wonder that there has been a quest for ways to reduce procrastination, or perhaps even to overcome it. In the following, we explain how some have tried to treat procrastination.

1.2. Procrastination interventions

Interventions in the field of procrastination all share the same aim—reducing the intention-action gap. However, they choose different paths to reduce this gap. The diversity of interventions (Van Eerde, 2015) mirrors the different theoretical perspectives that have been adopted in understanding procrastination (Klingsieck, 2013a). Schouwenburg (2004) offers the first, and as of yet only, attempt to categorize the interventional approaches. Based on a content analysis of interventions programs, he identifies three general themes (p. 198–199): 1) Training self-regulatory skills; 2) building self-efficacy; and 3) organizing social support.

The training of self-regulatory skills aims at establishing work habits that prevent procrastination. Typical modules of such trainings are stimulus-control techniques (in order the avoid being distracted from the target task), techniques of goal definition, and time management techniques (e.g., setting deadlines, defining time slots, monitoring progress). These techniques focus on changing the behavioral pattern of procrastination by using cognitive-behavioral interventions.

Second, the techniques in building self-efficacy focus on changing thinking patterns that sustain procrastination. Interventions in this category enhance self-efficacy by changing negative, unproductive, and inhibiting thoughts into positive, productive, and motivated thoughts.

The third category, peer support, aims at recognizing that others may deal with similar problems. As such, sharing these problems, collectively solving them, and reducing feelings of being the only one who suffers from these problems may help to reduce stress. In addition, peer support enhances social control mechanisms to monitor and remind others, thus helping to sustain the execution of intended behavior over time.

However, Schouwenburg (2004) overlooked some approaches such as the paradoxical approach that had been taken (e.g., Lopez & Wambach, 1982). Some of the interventions included in this categorization were atheoretical or combined different approaches. In addition, new approaches have been developed and combinations of these approaches have been implemented since 2004.

In general, the interventions can be thought of as more or less intense: Self-help books may be considered good advice, but the person needs to understand the materials and act upon them individually (Burka & Yuen, 2008; Ferrari, 2011; Pychyl, 2013; Steel, 2010a). A more intense type of treatment would be to have workshops or training sessions in which groups learn about procrastination and what may be strategies to change the behavior. An even more intense intervention would be a therapy, a guided approach with a therapist, mostly individual, but sometimes also administered in a group. In the following, we will systematize the interventions by presenting our meta-analysis.

1.3. The present study

The present study integrates procrastination intervention studies meta-analytically to assess whether, and if so to what extent, procrastination decreases after an intervention. It systematizes all interventions according to their characteristics (e.g., sample, intervention methods used). This systematization enables to compare studies in order to establish whether some types of interventions (e.g., therapy) caused larger effects than others. No distinction in the domain or context of procrastination was specified in advance, given that the intervention studies almost entirely focus on the domain of academic procrastination.

2. Methods

2.1. Literature search process

Both published and unpublished journal articles, book chapters, doctoral theses, and master theses were included in the meta-analysis. The literature was searched in the databases PsycINFO, ERIC, Web of Science, Scopus, and, Business Source Premier from 1806 to September 2016. For each database, slightly different search strategies were used. The search strategies can be found in the electronic supplement material (ESM1). An example of such a search based on keywords is the following: *procrastination (also: procras*, dilatory behavior*, postpon* behavior*) and systematic review (also: systematic review*), meta-analysis (also: meta-analy*), treatment outcome (also: treatment, treatment*), clinical trials, intervention (also: interv*, school based intervention, workplace intervention), therapy (also: therap*, online therapy), training (also: train*), program, counsel*, pretesting (also: pretest*, pre-test*), posttesting (also: posttest*, post-test*), treatment effectiveness, evaluation, random*, control group, control condition, quasiexperiment* (also: quasi-experiment*) or effect size**. After deduplication of the findings from the multiple databases, 989 results remained. The authors also hand searched the proceedings of the Biennial Procrastination Conferences (2009 until 2017) that have been held since 1999 by a group of interested researchers, and tracked citations backwards in Google Scholar (300 scanned, 69 selected). After inspecting these 989 publications on whether these actually concerned procrastination and possibly interventions, 324 publications remained.

2.2. Classification and selection

Both authors contributed in all steps of classification, selection, and coding. All information on all publications were organized in a database. In a first round of selecting, the titles and abstracts of all publications were screened to exclude all publications that did not meet the inclusion criteria. In order to be included in the meta-analysis, the study had to be about an intervention to reduce procrastination for a non-clinical population. Of the 324 studies screened, 89 publications remained. The other studies were either not about an intervention to reduce procrastination, or not about an intervention to reduce procrastination for a non-clinical population.

In a second round, the full text articles were read. Only publications that included a procrastination measure before and after the intervention, reporting means, standard deviations, and sample sizes were included. In the case of missing information, authors were contacted in order to receive the missing statistical information. This resulted in 24 publications to be included in the meta-analysis, containing 44 effect sizes pertaining to the pre-post measurement of procrastination, of which 35 employed an experimental or quasi experimental design in the pre-post assessment of procrastination (see Fig. 1). Eight of the 35 effect sizes came from unpublished sources (Binder, 2000; Broers, 2014; Moradi, 2015; Otermin-Cristeta, 2017).

Information on follow-up effects was also gathered, in order to establish whether effects remained stable after the study. The studies included 23 posttest-follow-up effect sizes, of which 12 contained comparisons to a control group.

2.3. Coding

To collect the relevant study characteristics from each publication of the experimental studies, a standardized form was used. Using this form, the following data were collected: (1) first author and publication year; (2) design (with or without control group); (3) type of control group (active or nonactive); (4) sample characteristics (student or other, and gender); (5) sample size; (6) characteristics of the intervention (type and duration) and (7) operationalization of procrastination as subjective or objective. Only studies that assessed procrastination with scales that had been shown to be reliable (e.g., Cronbach's Alpha higher than 0.70) and valid in previous studies were included. Among the subjective measurement scales we included trait and state measures: Subjective measures that conceptualize procrastination as a dispositional variable ask participants to rate how typical procrastination is for them. Examples of these scales are the General Procrastination Scale (GPS; Lay, 1986), the Irrational Procrastination Scale (IPS; Steel, 2010b), the Adult Inventory of Procrastination (AIP; McCown & Johnson, 1989), and the Pure Procrastination Scale (PPS; Steel, 2010b). Some of these scales specifically assess academic procrastination. Examples of these are the Aitken Procrastination Inventory (API; Aitken, 1982), Procrastination Assessment Scale-Students (PASS; Solomon & Rothblum, 1984), and the Tuckman Procrastination Scale (TPS; Tuckman, 1991). There was also one scale included that conceptualizes academic procrastination as a state: The Aitken Procrastination State Inventory (APSI; Schouwenburg, 1995) asks participants to rate the frequency of academic procrastination in the course of one week. Overall, the instruments have been shown to have low discriminant validity: They all measure procrastination but do not differentiate between different forms of procrastination such as decisional or arousal procrastination (Steel, 2010b). The electronic supplement material 2 (ESM 2) provides an overview of the coding schema in detail and all coded studies.

Some studies compared different treatments in experiments, such as different types of coaching (Losch, Traut-Mattausch, Mühlberger, & Jonas, 2016). In this case, we coded each treatment separately as if different experiments had been conducted. This means that the participants in the control conditions that have been compared to different treatments may be represented more than once. The random effects model we used allows for studies to be dependent because it makes no assumptions about the specific form of the sampling distributions of the effect sizes, nor does it require knowledge of the covariance structure of the dependent estimates (Hedges, Tipton &, Johnson, 2010). We did not distinguish randomized and non-randomized control groups, as only a small portion of the control groups were non-randomized (e.g., Van Eerde, 2003a, 2003b).

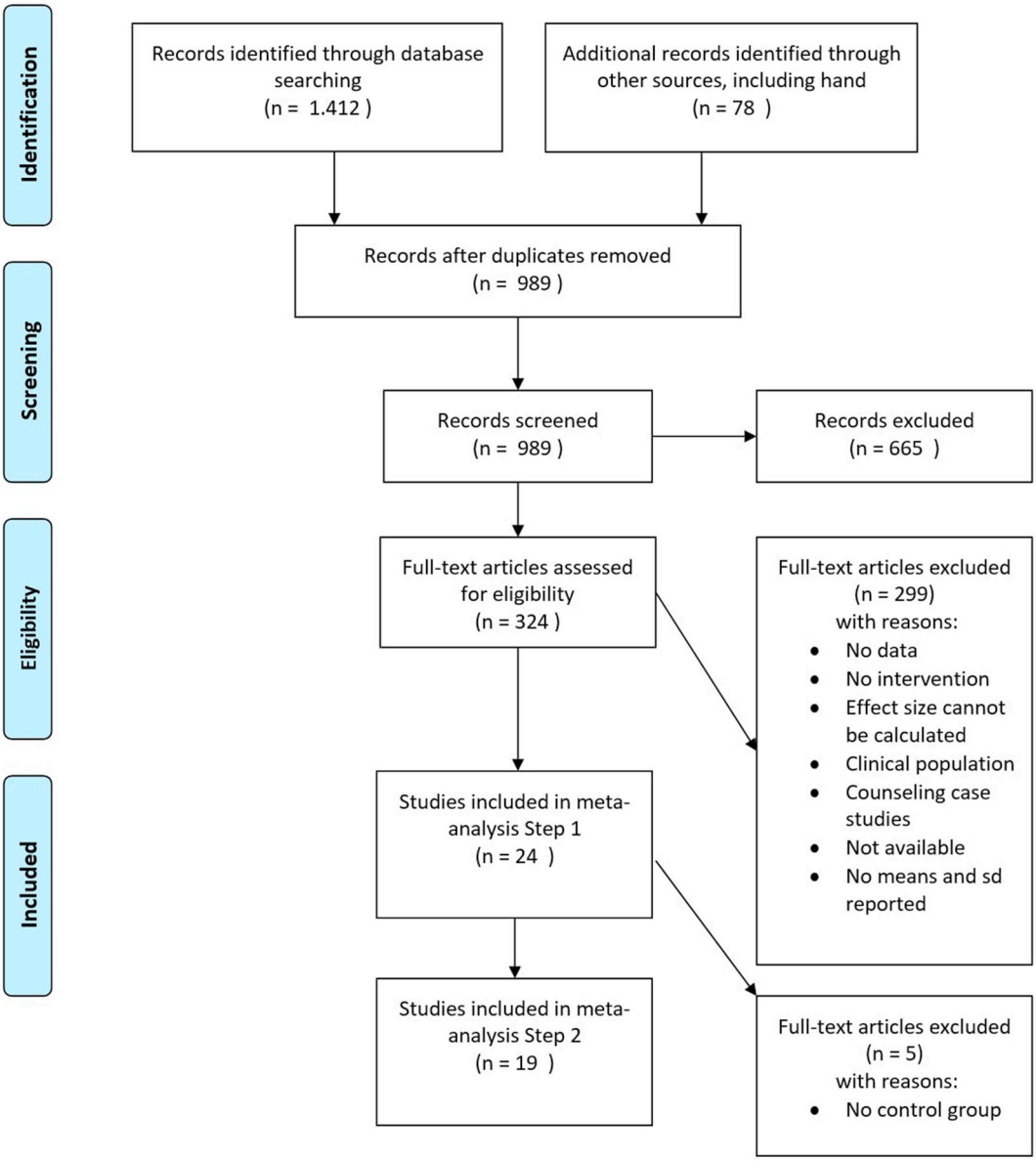


Fig. 1.

2.4. Statistical analysis

We distinguish two effects: (1) Pretest-posttest comparison. The intervention provides information on the change after treatment in comparison to before treatment; (2) Posttest-follow-up comparison. These effect sizes provide information on whether the effect remained stable, increased, or declined after completion of the intervention. We used a random effect model to calculate effect sizes and to conduct the moderator analyses.

To compute a pre-post effect size, a correlation coefficient is needed that represents the correlation between the repeated measures of procrastination. Following Lipsey and Wilson (2001), we took the test-retest correlation of the procrastination scale as reported in the research on the scale construction. If this information was not available, we used information from an intervention study that had included the within-subject correlation pre and post intervention, as $r = 0.76$ (Garrison, 2013). We report all effect sizes as the reduction in procrastination, with a negative sign indicative of a positive effect of the intervention in terms of lower procrastination after the intervention.

We used Del Re (2015) as a general source of information on meta-analysis procedures in R. We used the Metafor package to calculate the effect size (Viechtbauer, 2010), in particular the script that is based on Morris (2008) for computing a (standardized) effect size measure for pretest posttest control group designs. The Metafor package was also used to assess outliers (Viechtbauer &

Cheung, 2010) and publication bias, specifically drawing a funnel plot, conducting the Egger test (Egger, Smith, Schneider, & Minder, 1997) and Trim-and-Fill procedure.

2.4.1. Treatment moderators

The interventions in our datasets can be differentiated along certain characteristics concerning the interventions itself, such as the type and duration of the intervention. Other characteristics pertain to the evaluation of the intervention such as type of control group (waitlist versus active control group), the sample's characteristics (gender composition, student versus other sample type), and the publication year. Because these differences can be responsible for a high variability in overall effect sizes they were examined as moderators:

2.4.1.1. Type of intervention. We distinguished four types, initially based on Schouwenburg's (2004) characterization but recognizing new developments since 2004. We also included interventions that Schouwenburg (2004) overlooked, such as paradoxical approaches, and combinations of approaches. We distinguished the following types of procrastination interventions: 1) self-regulation, including time management and emotion regulation training; 2) cognitive behavioral therapy; 3) other therapeutic approaches such as paradoxical interventions, coherence therapy, and acceptance and commitment-therapy; and 4) new developments in the realm of concentrating on individuals' strengths and resources. (See the coding schema in the electronic supplement material 2 for the definition of the four categories.) Two research assistants who have a profound knowledge of procrastination literature rated all interventions according to these descriptions. Fleiss' Kappa between our ratings, the first rater's rating, and the second rater's rating was $\kappa = 0.64$ ($p < 0.001$). Cohen's Kappa was $\kappa = 0.70$ ($p < 0.001$) between our rating and the first rater's rating and $\kappa = 0.62$ ($p < 0.001$) between our rating and the second rater's rating. Cohen's Kappa for the ratings of the two raters was $\kappa = 0.63$ ($p < 0.001$). Thus, the interrater reliability was satisfactory (Fleiss & Cohen, 1973).

2.4.1.2. Self-regulation. Enhancement of self-regulation refers to the training of self-regulatory strategies (cf. Schouwenburg, 2004). Self-regulation encompassed the management of all internal (e.g., attention, vigilance, emotion, motivation, and volition) and external resources (e.g., work environment, social support, and time) when moving towards a goal. Typical aspects of enhancing self-regulation in procrastination interventions are self-reflection and -monitoring (Mühlberger & Traut-Mattausch, 2015), stimulus control (Grunschel, Patrzek, Klingsieck, & Fries, 2018), self-motivation (Green, 1982; Grunschel et al., 2018), and regulation of emotions (Eckert, Ebert, Lehr, Sieland, & Berking, 2016).

Time-management is a specific case of self-regulation. It is a "container concept that includes several tools to organize work and life in order to accomplish tasks effectively and efficiently" (Van Eerde, 2015, 312). It focuses on the use of time within organizing work and life. Typical aspects of time-management in procrastination interventions are goal setting (Gustavson & Miyake, 2017), planning, prioritizing, and organizing (Gieselmann & Pietrowsky, 2016; Van Eerde, 2003a, 2003b), and monitoring time (Broers, 2014). Because time management usually does not address any emotional issues and may not be sufficient to overcome procrastination (cf. Van Eerde, 2015), it is often combined with other approaches to sustain desired behaviors. Of all intervention studies reviewed, 13 relied on enhancement of self-regulation.

2.4.1.3. Cognitive-behavior therapy. Cognitive-behavior therapy (CBT) is a component of many procrastination interventions ($k = 12$). The underlying idea is that our thoughts determine how we feel and how we behave. Thus, the therapy focuses on identifying thoughts, checking whether thoughts are functional or dysfunctional, correcting dysfunctional thoughts, and transferring corrected thoughts into functional behavior. Aspects of cognitive-behavior therapy in procrastination interventions are exploring the personal experience with procrastination, understanding the personal pattern of procrastination, focusing on irrational thoughts, changing irrational thoughts into productive thinking, and changing the procrastination behavior (e.g., Ozer, Demir, & Ferrari, 2013). In some cases, time-management techniques such as prioritizing, goal setting, and, monitoring (Binder, 2000; Rozental, Forsell, Svensson, Andersson, & Carlbring, 2015b; Toker & Avci, 2015; Schubert Walker, 2004), and self-regulation techniques such as stimulus-control (Rozental et al., 2015b) are used for the transfer of corrected thoughts into behavior. However, the framework in which these techniques are applied is a cognitive-behavioral one.

2.4.1.4. Other therapeutic approaches. There are certain techniques that have been used in a CBT-framework but are also separately applied in procrastination interventions. These are paradoxical interventions, coherence therapy, and acceptance-based behavior therapy. Paradoxical interventions pertain to time-restriction techniques (Höcker, Engberding, Haferkamp, & Rist, 2012) or "do-not-study-but-procrastinate"-instructions (Lopez & Wambach, 1982; Wright & Strong, 1982). Coherence therapy aims at a deeper understanding of the emotional truth of a symptom (Rice, Neimeyer, & Taylor, 2011) while acceptance-based behavior therapy encourages procrastinators to view their painful thoughts and emotions as natural and transient. In a second step, they are then encouraged to define experiences that can direct behavior, rehearse the (new) positive thinking patterns, and to engage in valued activities (Wang, Zhou, Yu, Ran, Liu & Chen, 2017). We subsume all three approaches in the category "other therapeutic approaches" which included 7 interventions.

2.4.1.5. Strength and assertiveness. Very recently, another branch of interventions has emerged that cannot be allocated to any of the aforementioned three categories. This branch subsumes interventions on assertiveness training and strength-based trainings. Assertiveness or strength-based trainings aim to foster the self-efficacy of (young) procrastinators in order to prevent procrastination (Moradi, 2015; Visser, Schooneboom, & Korthagen, 2017). They do so by focusing on the resources and strengths of the procrastinators. In our sample, two procrastination interventions had this focus.

Besides these four general themes, procrastination interventions also included psychoeducation (e.g., Grunschel et al., 2018; Rozental et al., 2015a,b), social support (e.g., Binder, 2000; Schubert Walker, 2004; Van Eerde, 2003a, 2003b), and relaxation techniques (e.g., Binder, 2000).

2.4.1.6. Duration of intervention. We coded the total number of minutes spent in training or therapy.

2.4.1.7. Waitlist versus active control group. We distinguished the activities participants in the control groups engaged in. We coded for active (participants were involved in other activities assumed to rule out a justification of effort explanation for the participants) versus passive (mostly waitlist) control groups.

2.4.1.8. Publication year. We coded the year of publication of the study. As years go by, more effective interventions may have been designed, which would be visible in a positive relation between year and effect size.

2.4.1.9. Sample gender composition. We coded the percentage of women in the sample. Some studies indicated gender differences in procrastination, showing that men procrastinate more than women (Beutel et al., 2016; Steel, 2007; Van Eerde, 2003a, 2003b). This may affect the outcome of the intervention.

2.4.1.10. Student versus other sample type. We distinguished student samples from general samples.

2.4.1.11. Academic versus general procrastination scales[1]. There were at least nine different ways in the studies to assess procrastination. As the number of studies per scale was too low to compare effect sizes obtained with each of these scales separately, we then compared effect sizes obtained with all scales assessing academic procrastination ($k = 24$) with those that assessed procrastination using general scales (or that combined scales for measurement ($k = 10$).

3. Results

3.1. Pre-test post-test comparison

In this comparison, we included all studies that provided means and standard deviations of pre- and posttest, whether they used a control group or not ($k = 44$). We averaged all effect sizes. The average was found to be -1.07 [-1.32; -0.82], $p < 0.0001$. The effect sizes were found to be highly heterogeneous Q ($df = 43$) = 617.33, $p < 0.0001$, $I^2 = 94\%$. We then compared the treatments between the studies that had no control group ($k = 9$) with those that had a control group ($k = 35$). There was no significant difference between the effect sizes of both types of studies QM($df = 1$) = 0.33, $p = 0.29$.

Subsequently, we removed the effect sizes without control group and proceeded with only the studies that had control groups, as there are some problems with the interpretation of the effect sizes without comparison to a control group. That is, people may change over time, even without treatment, and as such, a control group helps to establish this general effect as a comparison for the intervention. Thus, in this analysis, we took into account the possible changes of control groups by making a comparison between the change in treatment and control groups. The overall effect of the 35 effect sizes showed a significant and a large effect in the changes between pretest and posttest of the treatment in comparison to control groups (-0.62 [-0.83; -0.41], $p < 0.001$). Fig. 2 shows the forest plot of the estimated effect per study. The sample showed considerable heterogeneity Q ($df = 34$) = 190.83, $p < 0.0001$, $I^2 = 82\%$. Subsequently, we inspected the effect sizes further, as one appeared to be extremely positive (see Fig. 2). The four types of diagnostics recommended by Viechtbauer and Cheung (2010) were plotted (see ESM 2), and these indicated that there was one outlier, namely the effect size found in the smartphone intervention for bedtime procrastination (Broers, 2014). After removing this outlier, the average of these 34 effect sizes in terms of the difference in standardized change between treatment and control group, was estimated as -0.65 [-0.85; -0.45], $p < 0.001$. However, there was still considerable heterogeneity after removing this effect size, Q ($df = 33$) = 162.52, $p < 0.0001$, $I^2 = 80\%$.

Furthermore, we checked for publication bias by drawing a funnel plot and conducting an Egger test for the asymmetry of the points in the plot (Egger et al., 1997). This test showed that the plot was asymmetrical ($Z = -3.03$, $p < 0.0024$). The trim and fill procedure revealed that 5 points were missing on the right side of the plot. If these points would be added, the effect size would only change slightly (-0.53 (se = 0.11) [-0.74; -0.32]). This provides some confidence that publication bias was not a severe threat. A comparison in effect sizes between the studies in our meta-analysis that were published ($k = 27$) versus unpublished ($k = 7$) was nonsignificant QM($df = 1$) = 0.24, $p = 0.62$.

3.2. Post-test follow-up comparison

In this analysis, we included studies that did and did not include a control group in the follow-up measurement, such that 23 effect sizes were combined. There was no significant change over time after the posttest (-0.05 [-0.16; 0.05]). The effect was heterogeneous Q ($df = 22$) = 64.10, $p < 0.0001$, $I^2 = 69\%$. Again, we inspected the effect sizes for outliers using the same procedure

[1] We thank the anonymous reviewer for addressing this distinction.

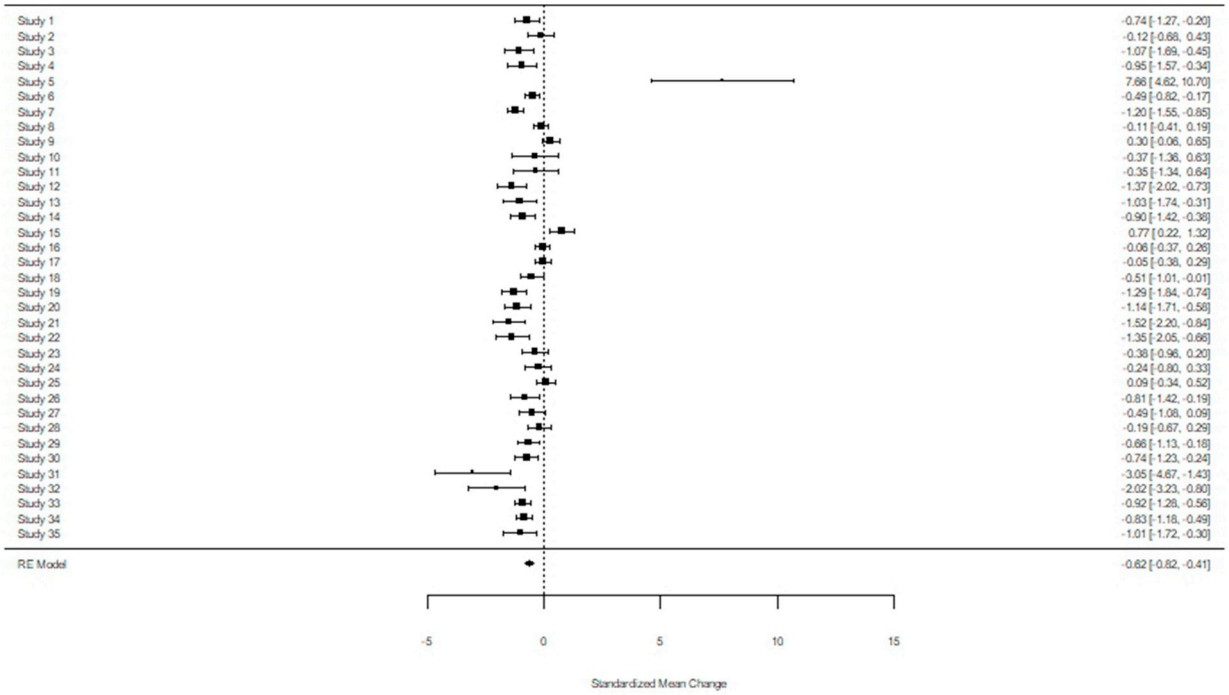


Fig. 2.

(Viechtbauer & Cheung, 2010). This revealed that the effect size found by Höcker et al. (2012) was an outlier, and we removed it. This caused a large drop in the heterogeneity of the effect sizes (now Q (df = 21) = 41.72, $p < 0.01$, I^2 = 49%) and a small but significant reduction in procrastination of −0.09 [-0.18; −0.00]. Subsequently, we tested whether the interval of the measurement between the posttest and the follow-up measurement moderated this effect size. The interval of the follow-up measurement varied between seven and 365 days. No moderating effect of the number of days was found QM(df = 1) = 0.22, p = 0.64. These results indicate that the effects of treatment remained stable and even slightly ameliorated after treatment, up to one year after the treatment.

We tested whether the 12 effect sizes from the studies using a control group in the follow-up phase differed from those that did not. There was no difference QM(df = 1) = 0.91, p = 0.34. We then proceeded to analyze the follow-up effect sizes obtained from the experimental studies.

The forest plot of the effect sizes of the twelve studies that had control groups in the follow up phase is provided in Fig. 3. The overall effect was non-significant, −0.01 [-0.16; 0.14], p = 0.79, indicating that procrastination remained stable between posttest and follow-up in comparison to a control group. The effect sizes were found to be homogeneous (Q (df = 1) = 7.08, I^2 = 0%), indicating that in all studies the effect of the intervention remained over time and did not differ from the control group. The length of the posttest–follow-up interval varied between seven and 182 days. The length of this interval did not affect the overall effect size QM (df = 1) = 0.29, p = 75.

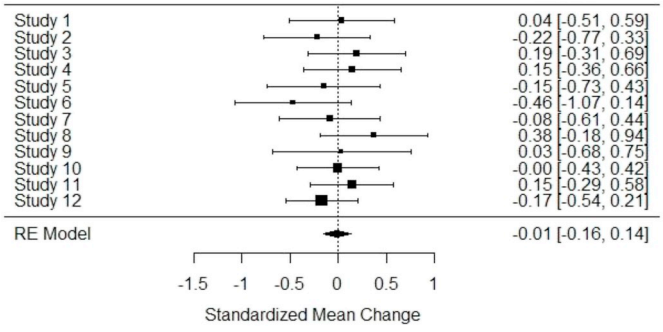


Fig. 3.

Table 1
Meta-regression of reduction of procrastination on moderating variables.

	B	SE	95% CI		*p*
Self-regulation	.24	.20	-.16	.64	0.24
Cognitive Behavior Therapy	-.45	.19	-.84	-.06	0.02*
Other Therapies	-.12	.26	-.63	.38	0.63
Strength and Assertiveness	1.04	.40	.26	1.82	0.001**
Duration	-.00	.11	-.21	.21	0.99
Waitlist vs Active Control Group	.36	.23	-.08	.80	0.11
Year of Publication	.03	.11	-.19	.25	0.79
Sample Gender	-.00	.00	-.00	.00	0.81
Student vs. Other	-.31	.26	-.81	.19	0.22
Academic vs. General Procrastination Scale	.30	.21	-.11	.71	0.15

Note. B = Regression coefficient; SE = standard error, CI = Confidence interval. *$p < 0.05$, **$p < 0.001$.

3.3. Treatment moderators

The high variability in overall effect sizes of the pre-post control effect sizes points at differences in underlying study characteristics that may serve as moderators. Table 1 provides the result of the meta-regression analysis.

3.3.1. Type of intervention

Each of the four types of interventions was compared. This analysis showed that there was no significant effect of studies coded as self-regulation ($k = 13$) QM(df = 1) = 1.40, $p = 0.24$; cognitive-behavior therapy ($k = 11$) was significant QM($df = 1$) = 5.11, $p = 0.02$, showing a larger reduction in procrastination, total effect −0.95 [-1.19; −0.72], also showing much less heterogeneity in effect sizes ($I^2 = 40\%$), albeit still significant Q ($df = 10$) = 16.99, $p = 0.07$; other therapies ($k = 8$) did not show a significant difference QM(df = 1) = 0.24, $p = 0.63$; and strength and assertiveness training ($k = 2$) had a significantly smaller reduction QM ($df = 1$) = 6.78, $p = 0.001$. The total effect of these two studies was nonsignificant (0.32 [-0.62 - 1.26], and the heterogeneity ($I^2 = 78\%$) was still significant Q ($df = 1$) = 4.61, $p = 0.03$, $p = 0.03$.

3.3.2. Duration

Subsequently, we tested whether the duration of the treatment, varying between 60 and 1440 min, $m = 404$; $sd = 365$) was related to the effect of the treatment. There was no effect of duration, QM($df = 1$) = 0.0003, $p = 0.99$, implying that the time spent on the treatment is not systematically related to the reduction in procrastination.

3.3.3. Waitlist control versus active control groups

We compared the effect sizes of studies employing a waitlist control group ($k = 26$) versus an active control group ($k = 8$). This effect was nonsignificant, QM($df = 1$) = 2.62, $p = 0.11$.

3.3.4. Year of publication

We also were interested whether the year of publication was related to the effectiveness of the treatment, which would imply that interventions have become more useful over the years. The range of years included was 1979–2017. There was no relation between the year and the outcome of the intervention, QM($df = 1$) = 0.07, $p = 0.79$.

3.3.5. Sample gender composition

There was no effect of the gender composition (range from 0 to 79% women in the sample) on the effect size of the intervention QM($df = 1$) = 0.06, $p = 0.81$.

3.3.6. Student versus other sample type

There was no effect of whether the intervention study had been conducted using a student ($k = 28$) versus other types of samples ($k = 6$) QM($df = 1$) = 1.49, $p = 0.22$.

3.3.7. Academic versus general procrastination scale

There was no difference between intervention studies that used a scale to measure academic procrastination ($k = 24$) versus a general scale or using a combination ($k = 10$) QM($df = 1$) = 2.05, $p = 0.15$.

4. Discussion

Based on our meta-analysis, we conclude that procrastination decreases after an intervention. The average effect size denotes a medium to large decline, indicating that treatment is worthwhile in general. Thus, the findings suggest that people can change. This is important for students who struggle with procrastination within the educational context, as self-regulation may be seen as an

important characteristic to successfully achieve academic goals. We also conclude that the follow-up measurements indicate that procrastination does not return to its level before training, nor declines more over time. Rather, the effect remains stable.

With regard to the type of treatment, we conclude that cognitive-behavior therapy shows larger effect sizes. Possibly, the larger effects of this therapy are due to the intensity of the intervention. Overall, it was difficult to assign labels to intervention types because these combined different approaches (e.g., psycho-education, self-regulation, and peer support all in one intervention). Thus, we consider the finding regarding the larger effect of therapy preliminary only, as more and better designed studies are needed, as we will explain below. The same caution applies to the finding about assertiveness and strength interventions, as in the study on assertiveness (Moradi, 2015) the control group was showing better results than the experimental group and only two effect sizes were in this category.

Focusing on the delivery of the treatment (e.g., duration), we show that these characteristics neither change the average effect sizes. Nor do we find any other effects that could reasonably be assumed to influence the outcomes, such as the year of publication, the type of control group, the type of procrastination scale or the gender composition of the sample.

The main shortcoming is that the number of effect sizes is rather low, providing limited possibilities to find any moderating factors. In addition, the lack of moderating effects may be due to the subjective assessments that were used for procrastination. Future research may benefit from procrastination rated by others. For example, in a study by Van Hooft (2014), coaches rated participants on procrastination. These ratings were better predictors of success in job search than the self-ratings of the participants. Van Eerde (2003a, 2003b) provided evidence of the convergence between self-reported procrastination of participants and how organized coworkers found them.

We conclude that a limited number of studies was found, and many of these did not qualify for inclusion because statistical information was missing. Except for some small studies on paradoxical therapy, randomized control designs have only been employed in the last few years. Most samples are small and considerable attrition rates can be observed. Some authors (Höcker, 2015; Höcker et al., 2012) have shown that dropouts and participants did not differ, but that correcting for missing values made effect sizes somewhat smaller, yet still large. However, for many of the studies, we could not find additional analyses about those who dropped out. This possibly provides a biased estimation of the effect sizes because it is possible that those who feel they do not benefit from the intervention may be those who drop out. A study on how people experienced one of the procrastination interventions (Rozental, Boettcher, Andersson, Schmidt, & Carlbring, 2015a) has shed more light on these issues. For example, some remarked that reading texts as part of the intervention was something they would procrastinate on. The motivating and perhaps fun characteristics of the intervention itself would be something to consider when it is being developed. The nature of the quantitative data rules out an in-depth assessment of the experience of the participants. However, these experiences might make the difference between a successful and a failed intervention for each participant. That is, factors such as rapport, motivation, and persistence could not be taken into account in this study but may be important in practice.

Our categorization of interventions may be criticized, as we strayed from the theoretical categorization as suggested by Schouwenburg (2004) based upon new developments. However, we feel that there is value in covering the complete spectrum of approaches rather than, for example, removing therapeutic approaches as irrelevant to the educational context, as these have been applied successfully in the academic domain. Concerning the subjective measures of procrastination that were implemented in most studies, we point to Steel's (2010a, 2010b) critique that some of them are not one-dimensional, and of Haghbin (2015) that most of them have not been developed with the precise definition of procrastination in mind. However, we included studies in this review regardless of which measure of procrastination they used. In future evaluation studies, it might be worthwhile to combine the subjective assessment with some objective proxy of procrastination in the form of delay (such as assignment submission time, Miyake, Carruth, Gustavson, Lurquin, & Kane, 2017), or a peer rating type of measurement of how severe the procrastination appears to be to others.

One observation that can be made about the theoretical background of the original studies is that the procrastination literature hardly considers self-determination theory as a theoretical lens to understand nor as a starting point to diagnose and reduce procrastination. On the contrary, most of the studies conceptualize procrastination as a failure of self-regulation defined by the social cognitive approach. This not only explains why the cognitive-behavioral therapy is the most effective one in reducing procrastination but also calls for future intervention studies based on self-determination theory. Encouraging active students' initiatives would be one of the ways in which self-determination can be enhanced and as such serve as a remedy.

Other points that need to be addressed in future evaluation of procrastination interventions are (cf. Graham & Harris, 2014): (1) rigorous designs and comparable interventions such as active control group experiments (randomized control trials) and studies with longer follow-up periods (e.g., Rozental, Forsell, Svensson, Andersson, & Carlbring, 2017; Visser, Schoonenboom, & Korthagen, 2017), to find out more about the longer term effects of interventions; (2) credible control conditions that are as similar as possible to the tested intervention—including the time invested—but without the component theorized to affect procrastination; (3) construct valid dependent variables that not only cover the irrational delay but also the *suffering* due to the delay (cf. Haghbin, 2015). If possible, an objective measure of delay would help to show how internal norms and observed behaviors correspond; (4) a multi-faceted assessment of interventions' effects that address in depth how participants experienced the intervention (cf. Rozental et al., 2015a). In some cases, anxiety or stress-related measures were taken, or performance was measured in terms of grades. These could provide additional information on the intervention and a wider view of the importance of procrastination for other outcomes; (5) properly powered evaluation studies. Future studies should determine the number of participants needed in order to conduct an adequately powered study, keeping the problem of attrition in mind and countered by carefully crafted incentives; (6) transparent sampling procedures that include a discussion of how participants were included in the study and, thus, how they came about being labeled as procrastinators (for ideas cf. Engberding, Frings, Höcker, Wolf, & Rist, 2011; Rozental et al., 2015b). Also, in terms of

sampling periods, careful consideration should be given to which intervals need to be included and whether the effects of an intervention may be sustained over time. This may include additional measurements to the pre-test post-test follow-up designs, such as time series or diary studies (cf. Schmitz & Wiese, 2006).

In terms of guidelines for future interventions we conclude that the studies were not only very heterogeneous in terms of research designs but also in terms of content. Whereas some are based on theories (e.g., Gieselmann & Pietrowsky, 2016; Grunschel et al., 2018), others vaguely relied on empirical findings concerning procrastination (e.g., Ozer et al., 2013). In future studies, theory-based and evidence-based interventions would be preferable, for example, based on goal setting as well as cognitive-motivational and emotional self-regulation principles carefully juxtaposed and systematically compared in a series of studies. In addition, web-based and computerized deliveries of interventions have two potential advantages to face-to-face delivery: first, the possibility to match the treatment with the individual needs of the participant by using adaptive release of relevant materials; and second, the possibility to reduce costs by improving the efficiency of treatments of those who guide them and the participants. An overview of computerized therapies (Budney, Marsch, & Bickel, 2015) has provided information on psychosocial interventions. Although these concern the treatment of substance addiction—possibly having much more severe consequences than procrastination—the general notions are useful for future interventions on procrastination. These authors have described motivation enhancement, cognitive-behavioral therapy, community reinforcement approach, and contingency management as four types of interventions that have been tested in computer mediated treatment. In addition, SMS or other prompting, virtual reality, digital social support networks, and video games for teaching skills may also be used (cf. SuperBetter; Roepke et al., 2015). Such types of smartphone applications, based on gamification principles, have also been shown to be suitable for the treatment of procrastination (Klingsieck et al., 2017).

5. Conclusion

This study provided a meta-analytic integration of effect sizes obtained from procrastination intervention studies. The average effect size indicates that substantial reduction on procrastination may be achieved, and cognitive-behavior therapeutic approaches were shown to have larger effects. However, we also pointed out the limited number of studies, and possibly as a consequence, the lack of moderating effects. We hope that our study provides a useful overview and fruitful guidelines for the developers and evaluators of new interventions to the benefit of those who suffer from procrastination.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.edurev.2018.09.002.

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Further reading

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